# Data Analytics Internship Portfolio

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**Track:** Data Analytics Internship @ DecodeLabs

## Week 3

### Project 3: SQL Data Analysis & Business Intelligence

SQL | Relational Logic | Query Optimization | Aggregations

The final extraction phase transitioned from visual analysis to declarative relational querying. The queries were engineered following the strict database execution order (FROM → WHERE → GROUP BY → HAVING → SELECT → ORDER BY) to ensure optimal computational efficiency and avoid common syntax traps.

#### 1. High-Value Transaction Filtration (VIP Orders)

Engineered declarative `SELECT` and `WHERE` clauses to apply a strict row-level funnel on the dataset.

> ***Analytical Insight:*** *By filtering transactions exceeding the $1000 threshold and applying* `ORDER BY DESC`*, the query instantly isolated maximum revenue drivers, providing actionable targets for premium account management and VIP retention strategies.*

#### 2. Categorical Performance & Revenue Aggregation

Utilized `GROUP BY` alongside aggregate functions like `SUM()` and `COUNT()` to collapse raw rows into distinct product buckets.

> ***Analytical Insight:*** *Tracking both transaction frequency (Order Count) and total financial health (Revenue Sum) per category created a comprehensive structural map of demand, revealing which products form the financial backbone of the enterprise.*

#### 3. Premium Product Analysis & Post-Aggregation Filtration

Deployed the `HAVING` clause to filter grouped buckets based on computed averages (AVG).

> ***Analytical Insight:*** *Successfully bypassed execution order limitations (the Alias Trap) to isolate premium catalog items where the average unit price exceeded $150. This establishes clear boundaries between standard and premium product tiers.*

### 4. Revenue Percentage Contribution Modeling

Integrated subqueries with mathematical computations within the SELECT statement to calculate exact market share.

> ***Analytical Insight:*** *Computing the exact percentage contribution of each product relative to the total overarching revenue provided a transparent, executive-level view of inventory dependency and risk distribution.*